Guilhem Marion | Post-doc in Cognitive Neuroscience of Music

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Education

PhD Thesis in Neuroscience of Musical Preferences **Paris, France**
École Normale Supérieure (ENS) *2019–2023*

Title :Why do we Like (the) Music (we Like) ?

Cognitive Science : Protocol and stimuli design and analysis for EEG, behavior, fMRI, animal model, sociology and genetic twin design studies.

Neuroscience : Analysis of M/EEG, ECoG, intra-cranial in ferrets and birds, FUS and fMRI data using event-based, regression and classification algorithms.

Computer Science : Development of two new statistical models of music (Markov chains and bayesian based) and design of neural networks for modeling cognitive data in Python and Matlab.

Sociology : Collection of sociology field data in Rome and Paris including semi-directed interviews.

ATIAM **Paris, France**
Institut de Recherche et de Coordination Acoustique/Musique (IRCAM) *2018–2019*

Computer Science : Computer Music, MIR, Machine Learning, Music Analysis, Real-Time.

Signal Processing : Signal Analysis, Transformations, Synthesis, Reconstruction and Denoising, Source Separation.

Acoustics : Acoustics of musical instruments, Modal Analysis, Vibrations, Room Acoustics.

Coding practice in Python, C and Matlab.

Master's Degree in Computer Science **Lyon, France**
University Claude Bernard *2017–2018*

— Artificial Intelligence (from prolog to RNN), Natural Language Processing, Signal Processing, Web, NoSQL databases.
— Coding practice in Java, C++, Python, JS.

Preparing DEM Music Diploma **Lyon, France**
Music Conservatory *2016–2018*

Jazz guitar, and collective practice.

Electroacoustic Composition and Classical and Jazz harmony.

Master's Degree in Musicology **Lyon, France**
University Lyon Lumière *2016–2017*

— Achievement of a thesis about computer-aided methods for grammar modelization in a semiological approach to music.
— Courses in history, aesthetics, and sociology of music.
— Wrote of many essays for classes validation.

Bachelor's Degree in Theoretical Computer Science and Mathematics **Lyon, France**
École Normale Supérieure *2015–2016*

Theoretical Computer Science Advanced Algorithms, cover on the year of all notions from *Introduction to Algorithms* by Cormen and al., Graph Theory, Turing Machines, Church thesis, Formal Languages, Grammars, Computability, Complexity, System, Networks.

Mathematics Probability : Sample Space, Conditional Probability, Independence, Continuous Random Variables, Concentration Inequality, Limit Theorems, Discrete Markov Chains, Statistics. Logic.

Programming in C, C++, Python, Bash, and assembler for several projects. Design of a processor, Oriented Object Programming, Algorithms Optimization and GPU Programming.

Complementary courses in Film, Art History, Philosophy, and Linguistics.

Double Degree in Computer Science and Biology Ecology — **Montpellier, France**
University of Montpellier, 2^{nd} year of bachelor — *2013–2015*

Computer Science Algorithms, Databases, Statistics, Imperative Programming, Oriented Object Programming. Coding practice in C++, Java, Python, Bash and, Oz.

Biology Concepts and methods for Ecology, Mathematics and Statistics applied to Biology, Evolution, Genetics, Plant and Animal Physiology. Coding practice in R.

Experiences

Internship in Music Cognition — **Paris, France**
LSP, École Normale Supérieure — *February 2019 – July 2019*

Design of an EEG experiment about *audiation* in music.

Extending the work of Marcus Pearce on modelling melodic expectation.

Confroncting a new model of melodic expectation with EEG data.

Internship in Automatic Music Generation — **Lyon, France**
LIRIS, University of Lyon — *November 2017 – August 2018*

Bibliographic review about generative neural network and music generation.

Design of a neural network from a state-of-the-art knowledge (bi-directional LSTM) to generate jazz standards from the RealBook dataset, including melody and chords progression.

Work about memory in several musical grammar learning model.

Internship in Ethnomusicology (for Musicology Master's degree) — **Chișinău, Moldova**
AIRM, Moldavian Instition for Cultural Conservation — *Summer 2017*
funded by région Rhône-Alpes

Autonomously collected elements from sacred musical heritage : recordings, library, interviews, field work.

Partnership work with the AIRM about the enhancement of the Moldovan cultural heritage.

Internship in Computer Science with Alexander Shen — **Montpellier, France**
LIRMM, Team Escape, University of Montpellier — *Summer 2016*

Theoretical considerations on random and pseudo-random numbers.

Bibliographic review on production and test of random sequences.

Design of a random number generator about cryptographic quality.

President of a Film Association — **Lyon, France**
Champ Libre ENS de Lyon — *2015 – 2016*

Management of projections, debates and inviting filmmakers.

Internship in Computer Science with Sabrina Ouazzani — **Montpellier, France**
LIRMM, Team Escape, University of Montpellier — *Summer 2015*

Bibliographic review : Computability, Logic and Turing Machines.

Computer Music project using constraint modelling to generate counterpoint.

Peer-Reviewed Publications

1. **Guilhem Marion**, Camille Barbarot, Rupesh Kumar Chillale, Claire Pelofi, Shihab Shamma (in prep), *Learning To Enjoy Music : The Neural Underpinnings of Musical Enculturation*.
2. **Guilhem Marion***, Claire Pelofi*, Pablo Ripolles, Giovanni Di Liberto, Shihab Shamma (in review), *Cross-cultural Perspectives on the Predictive Coding Perspective of Music Perception*, Trends in Cognitive Sciences.
3. **Guilhem Marion**, Giovanni Di Liberto, Shihab Shamma (in review), *IDyOMpy : a Python Implementation for IDyOM, a Statistical Model of Musical Expectations*, Journal of Neuroscience Methods.
4. Yashish M. Siriwardena, **Guilhem Marion**, Shihab Shamma (2022), *The Mirrornet : Learning Audio Synthesizer Controls Inspired by Sensorimotor Interaction*
5. ***Guilhem Marion**, Giovanni Di Liberto, Shihab Shamma (2021), The music of Silence. Part I : Responses to Musical Imagery Encode Melodic Expectations and Acoustics, JNeurosci.*
6. Giovanni Di Liberto, **Guilhem Marion**, Shihab Shamma (2021), *The music of silence. Part II : Cortical Predictions during Silent Musical Intervals*, JNeurosci.
7. Giovanni Di Liberto, **Guilhem Marion**, Shihab Shamma, (2021), *Accurate Decoding of Imagined and Heard Melodies*, Frontiers in Neuroscience.
8. Shihab Shamma, Prachi Patel, Shoutik Mukherjee, **Guilhem Marion**, Bahar Khalighinejad, Cong Han, Jose Herrero, Stephan Bickel, Ashesh Mehta, Nima Mesgarani (2020), *Learning Speech Production and Perception through Sensorimotor Interactions*, Cerebral Cortex.

Talks

1. *Musical Enculturation : Toward The Neural Substrates of Individual Differences in Musical Enjoyment*, Invited speaker for the Department of Brain & Behavioral Sciences, Pavia, Italy, June 14, 2023.
2. *Neural Underpinnings of Musical Enculturation* , Talk for the Neuro Platform, Laboratoire des Systèmes Perceptifs, ENS, Paris, France, March 1, 2023.
3. *Are Expectation Mechanisms shared across senses ?* , Invited speaker for the Workshop on Aesthetic Values and Decision-Making in the Brain , Institut d'études avancées de Paris, France, October 22, 2022.
4. *Structures in Music, Structures in the Brain : Toward a New Definition of Musical Culture*, Invited speaker for the Neuropsychology and Imaging of Human Memory group, Caen University, France, October 3, 2022.
5. *Structures in Music, Structures in the Brain : Toward a New Definition of Musical Beauty*, Invited speaker for a symposium on Beauty studies, Cerisy Castle, July 18, 2022.
6. *Structures in Music, Structures in the Brain : Toward a New Definition of Musical Culture*, Invited speaker for a study-day about Cognition and Music, CREAA Strasbourg University, May 13, 2022.
7. *Investigating Cortical Correlates of High-Level features of Natural Speech and Music*, CNS, Symposium Speaker, San Francisco, April 26, 2022.
8. *Musical Enculturation Through Passive Exposure to Unfamiliar Music*, online, Invited speaker for CLAME Lab Meeting, 2022.
9. *The Music of Silence : Decoding Imagined Music and Inter-Notes Silences Through EEG*, online, Invited speaker for CLAME Lab Meeting, 2021.

10. *Neural Underpinnings of Musical Imagery*, Invited speaker for CogHear workshop, online, June 11, 2020.
11. *Neural Underpinnings of Musical Imagery*, NYU, Invited speaker for Peoppel's Lab Lab Meeting, 2019.
12. *Neural Underpinnings of Musical Imagery*, University of Maryland, Washington DC, Invited speaker for NSL Lab Meeting, 2019.

Book Chapters and General Public Publications

1. Intervention for the radio show Affaires Culturelles on the French public radio **France Culture** about musical preferences, June 2023.
2. Clélia Zernik and Justin Jaricot, ABÉCÉDAIRE DE LA BEAUTÉ, **book chapter** : *Norms in Music and Face Perception : How Does the Sense of Beauty Emerge*, B42 Editions, 2022.
3. *Why certain types of music make our brains sing, and others don't*, **The Conversation**, in English, French and Spanish, 220k readers, 2022.
4. Participation for a youtube video of the french channel **Spline LND**, *Can E.T. listen to Bach ?*, 12k views, 2022.
5. *How Does the Brain Represent Musical Culture*, **Forum des sciences cognitives** Study day for the general public, Paris, April 3, 2022.
6. Multiple interviews about the article The Music of Silence (JNeurosci) for multiple French and International media such as : **Science et Vie**, Cordis (EU), UMD News, Technology Networks, Science Daily, Medical News Today, Inverse, TCD, Neuroscience News, News-Medical, Scitech Daily, Florida News Times, The Health News Express, Mixpoint, Ani News, New Kerala, Eurekalert, 2021.
7. Talk for **La Semaine du Cerveau** (National French event about neuroscience for the general public) about musical cultures and the brain, March 2021.
8. Guilhem Marion & Jackson Graves, Participation to the radio show **Métaclassique** about musical expectations, 2020.
9. Participation to European Research Night (2020), Fête de la science (2020, 2023), Cognivence (2022), and high school interventions.

Grant Awards

1. Co-writer of the NSF grant : *The computational and neural basis of statistical learning during musical enculturation*, Collaborative NSF grant between NYU and UMD (USA), 900k, 2023.
2. Fyssen Post-Doc grant to study at the Peoppel's Lab in NYU (USA), 45k, 2023-2024.
3. École Française de Rome Travel Award (ITA), Hosting in Rome, and support by the EFR + 1k research money, 2023.
4. Neurotech Travel Award for the Telluride Neuromorphic Cognition Engineering Workshop (USA), 5k , 2022.
5. Chaire Beauté(s) PSL-L'Oréal (FR), 3 years PhD funding + 15k of research money, 2020.
6. Neurotech Travel Award for the Telluride Neuromorphic Cognition Engineering Workshop (USA), 5k , 2019.

Supervision

1. Fei Gao (master), Spectrogram-based Statistical Models of Music : From Bayesian Inferences to Transformer-based Deep Neural Networks, remote, September-June (10 month, part-time) 2024-2025.
2. Henri Le Goff (bachelor), Toward a New Methodology for a Socio-Psychology of Musical Preferences, ENS, Paris, France, 2022-2023 (10 month, part-time).
3. Agathe Mangialomini (master), Litterature Review of the factors Influencing Musical Preferences, ENS, Paris, France, 2022-2023 (6 month, part-time).
4. Hugo Dubois (master), The Neuroscience of Music (CNSM musician student), ENS, Paris, France, 2022 (6 month, part-time).
5. Amélie Picard (bachelor), A New Beyasian Model for Musical Expectation, ENS, Paris, France, 2021-2022 (6 month, full-time).
6. Camille Barbarot (bachelor), Neural Underpinnings of Musical Enculturation (EEG), ENS, Paris, France, 2021 (6 month, full-time).
7. Mai-An Nguyen (bachelor), co-supervised with Jackson Graves, Artificial Creation of Ambigious Sound Samples of Spoken Words (Laurel/Yanny-like), ENS, Paris, France, summer (1 month, full-time) 2020.

Posters

1. **Guilhem Marion**, Giovanni Di Liberto, Shihab Shamma, *Melodic Expectations are Encoded in the Brain during Musical Imagery*, ARO 2021.
2. **Guilhem Marion***, Giovanni Di Liberto*, Shihab Shamma, *The music of silence reveals neural auditory prediction*, APAN 2020.

Skills

Programming: Python, Matlab, C, C++, PyTorch, numpy, R, Bash, Java, JavaScript

Desktop tools: LaTeX, Illustrator, Indesign, HTML5, CSS on Linux and MacOS

French: Fluent.

English: Fluent.

Italian: At least B2 level.

Espagnol: Intermediate level (B1).

Art: Ableton Live, Pro Tools, MuseScore, Adobe Première Pro, Photoshop

Activities

Art Video: Exhibition at Taipei Fine Arts Museum (2019/04/13 - 2019/07/14).

Music: Multiple concerts in Jazz bands in Paris, Montpellier and Lyon.

Photography: Analog and digital photography. Exhibitions and publications for Art journals.

Climbing: Regular practice indoor and outdoor. 7a+ level in route and 7a in bouldering. Work in progress on *Sharma Sutra* 7a+ bouldering (V7) at prieuré, Lodève.